

NASA Kepler Exoplanet Classification Project Report

Kepler Exoplanet Classification Using Machine Learning

Abstract

I worked on a project where I used machine learning to classify Kepler Objects of Interest into three categories: CONFIRMED, CANDIDATE and FALSE POSITIVE. I tried out a few machine learning models like Random Forest, XGBoost, CatBoost and LightGBM. After comparing how well they did I chose LightGBM because it gave the results overall. I also used SHAP to see how the model made its predictions. This project taught me a lot about the machine learning process from cleaning up the data to explaining what the model predicted.

When the NASA Kepler telescope looks at stars it finds changes in brightness that might mean a planet is passing in front of the star.. Not all of these changes are actually caused by a planet. Some are noise or other things that happen in space.

My goal was to build a machine learning model that could automatically sort these changes into three groups: CONFIRMED, CANDIDATE and FALSE POSITIVE. I wanted to know which model worked best and why it made the predictions it did.

Dataset Overview

I used the NASA Kepler KOI dataset for this project. The dataset has information about things like the time it takes a planet to go around a star, how much the stars brightness changes when a planet passes in front of it the size of the planet and the temperature of the planet.

The thing I was trying to predict was the koi_disposition, which's the final classification for each object.

Data Preparation

First I looked at the dataset to see if there were any values, duplicate rows or columns that were not helpful. I got rid of the columns that did not help the model learn and filled in the missing numbers with the value because it is not affected as much by outliers.

After I cleaned up the data I changed the target labels into numbers. Split the dataset into two groups: one for training and one for testing. I used 85% of the data for training. 15% For testing.

Model Development

I tried out four models:

- * Random Forest
- * XGBoost
- * CatBoost
- * LightGBM

I tested each model using Stratified 5 Fold Cross Validation so it would be fair. LightGBM did the best. I chose it as my final model.

I also tried out settings for the model and compared the new model to the original one before choosing the final version.

Model Evaluation

I tested the model using **accuracy, precision, recall, F1 score and a confusion matrix**.

The model did well on the **FALSE POSITIVE** group and also did well on the **CONFIRMED** and **CANDIDATE** groups. Most of the mistakes happened between CONFIRMED and CANDIDATE because these two groups are similar.

Explainability

I used SHAP to see which features had the impact on the model.

The SHAP plots showed that things like the time it takes a planet to go around a star and other features related to the Kepler telescope were important for making predictions. Using SHAP helped me understand that the model was making decisions based on information not just random patterns.

What I Learned

Before I started this project I did not know much about machine learning.. During this project I learned how to clean up data compare different machine learning models test their performance and explain what the models predicted using SHAP.

I also learned that just getting an accuracy is not enough. It is also important to understand why the model makes its predictions.

Limitations

Even though the model did well it still got some things wrong between the CONFIRMED and CANDIDATE groups. I also found that some features were very influential. I checked them carefully to make sure they were not affecting the results in a bad way.

Future Work

If I keep working on this project I would like to try out some things like adding more features combining models and using more advanced ways to optimize the models settings. I would also like to test the model on datasets of exoplanets.

This project gave me hands-on experience, with the machine learning process. I built a model that can classify Kepler exoplanet candidates well and can explain why it makes its predictions. Importantly I learned how to start and finish a machine learning project and explain the results in a simple way.